

INSTALLATION GUIDE

Enterprise Server Deployment and Operating System Installation

ITEP 414 – System Administration and Maintenance | Week 3

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Company Scenario: ABC Startup Solutions

1. Project Overview

This guide documents deployment of a clean Ubuntu Server LTS virtual machine for ABC Startup Solutions. The server provides a foundation for future file sharing, remote administration, database hosting, web hosting, and internal services.

2. Objectives

- Install Ubuntu Server LTS in a virtual machine.
- Configure hostname, user, storage, networking, and SSH.
- Verify login, hostname, IP, internet connectivity, updates, and SSH.
- Produce reproducible technical documentation.

3. Software Requirements

Software / Media	Purpose
Oracle VirtualBox or VMware Workstation	Virtualization platform
Ubuntu Server LTS ISO	Linux server installation media
Windows Server Evaluation ISO	Bring-home installation/comparison activity

4. Virtual Machine Specifications

Component	Configuration
Name	Ubuntu-Server-Week03
RAM	4 Gb
CPU	2 virtual processors
Storage	40 GB VDI/VMDK
Network	NAT or Bridged if instructed
Optical Drive	Ubuntu Server LTS ISO
Hostname	server01
SSH	OpenSSH Server enabled

5. Step-by-Step Ubuntu Server Installation

- 1). Create the VM. Use 4 GB RAM, 2 vCPUs, and a 40 GB virtual disk.
- 2). Attach the ISO. Attach the Ubuntu Server LTS ISO to the virtual optical drive.
- 3). Language and keyboard. Select English and the appropriate keyboard layout.
- 4). Network. Accept DHCP unless instructed otherwise.
- 5). Hostname. Set the hostname to server01.
- 6). User. Create a non-root administrative user with a strong password.
- 7). Storage. Select Guided – Use Entire Disk and document the partition scheme/filesystem.
- 8). SSH. Select Install OpenSSH Server.
- 9). Finish. Complete installation, reboot, and remove/eject the ISO if prompted.

6. Initial Server Configuration

hostname

ip addr

sudo apt update

sudo apt upgrade -y

Record the actual assigned IP address: [ENTER YOUR IP ADDRESS].

7. Verification Procedures

Test	Command / Action	Expected Result
Login	Log in with created user	Successful shell login
Hostname	hostname	server01
IP	ip addr	Valid interface/IP
Internet	ping -c 4 google.com	Successful replies
Updates	sudo apt update && sudo apt upgrade -y	Updates complete
SSH	systemctl status ssh	active (running)

8. Screenshot Evidence

01-login.png

02-hostname.png

03-ip-address.png

04-ping.png

05-system-update.png

06-ssh-status.png

Store screenshots in Week03/screenshots/.

9. Troubleshooting

Problem	Likely Cause	Action
No IP	Adapter/DHCP	Check VirtualBox network mode and DHCP
Ping fails	Routing/DNS	Check ip addr, route, DNS and network mode.
SSH inactive	Service missing/stopped	Install openssh-server and enable/start ssh
Boot failure	Disk/ISO/boot order	Check virtual disk and remove ISO.
sudo failure	User permissions	Use an authorized admin account.

10. Best Practices

Use least privilege, strong passwords, regular updates, SSH controls, firewall rules, backups, and change documentation. Never commit passwords or private keys to GitHub.